ATS C Sample initial release precondition

C Sample Drawing initial Release Checklist

C Sample Drawing Release Pre-condition Checklist		Result/Report	Status	Resp.	Date
Design	1. C Sample 3D model confirmed by customer?				
	2. C Sample offer drawing check with manufacturing drawing?		OK	卢青松	2026.3.9
	3. C Sample Nameplate confirmed by customer? Email or drawing confirm?				
	4. D-FMEA first version finished?				
	5. I-FMEA first version finished?		Covered by 493	卢青松	2026.3.9
	6. Installation guideline finished?				
	7. PSC/CC list finished for CMK summary?		已有初版	卢青松	2026.3.9
PUR/MOE	1. C Sample Make and Buy confirmed?				
	2. C sample DFMA welding process confirmed with MOE? Flow chart?		ongoing	[illegible]	
	3. P-FMEA first version plan?				
DV Validation	1. Canning validation finished ? report?		基于FEA，风险较低 Covered by 主项目	杨广松	2026.03.10
	2. ATS DV validation finished? report?				
System	1. TCD status?		covered by 493	沈伟波	2026.3.25
	2. POC result & report ?				

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BOSCH 博世 Confidential	Product Development Design Change Request 产品开发设计变更请求	Version 15 Effective date:2026-01-14th Design Change Request No. 26_001

Customer project Name/ 客户项目名称:	JIM-PT611	MCR No. / MCR号:	
Sample status / 样件状态:	☐ A Sample ☐ B Sample ☑ C Sample ☐ D Sample		

Change part&product Name/更改零部件产品的名称(简要描述)

DOC+SDPF UNIT: F01ZH003RX-02 SCR UNIT: F01ZH003VH-00 BRACKET: F01Z5103J8-00

Reason of changes/更改理由 (简要描述):

按客户要求释放JIE PT611 B样件;

Change from/变更来源:		
Customer request / 客户要求 ☑	Supplier request / 供应商请求 ☑	Design optimization / 设计优化 ☑
Correction / 更正(错误) ☐	Other / 其他 ☐ ( )	

Change inform to/变更通知人:	Zhang Shengchang,Zhang Hanquan,Shen Weibo, Li Yicen, Ge Jilong Yang Guangtuo,Jia Kebin, Luo jing, Lu Qingsong, Min Lingyun

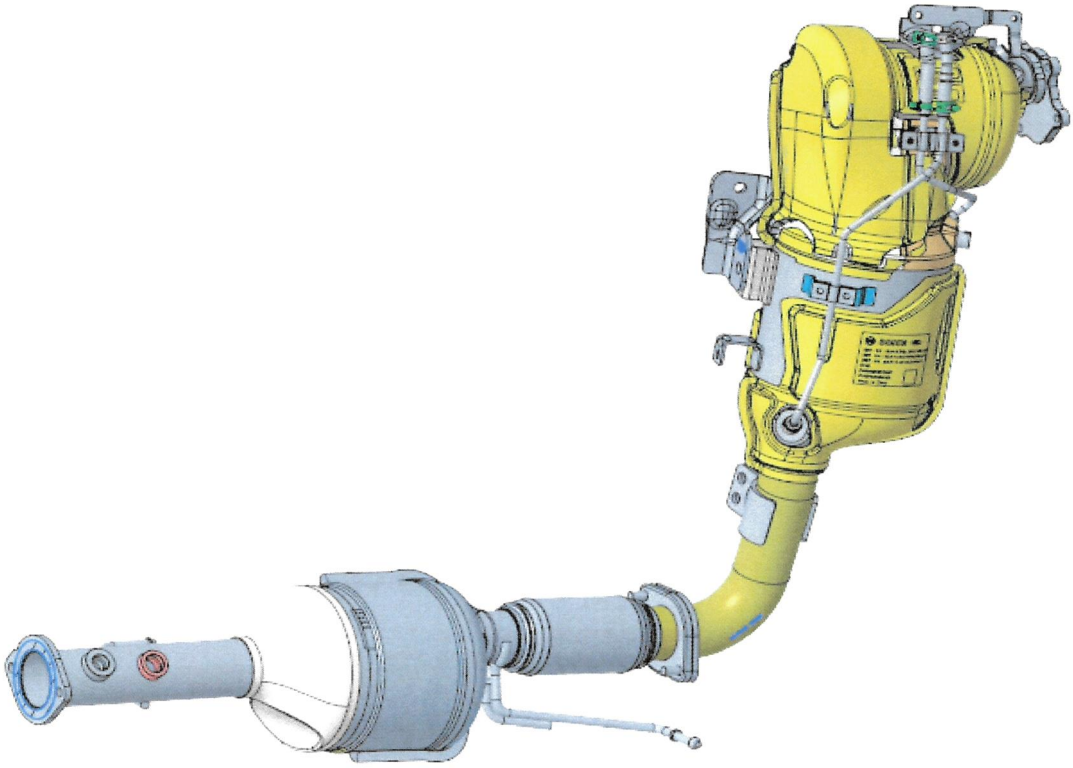
Whether this change impact others component or product 此变更是否影响到其他部件和产品: ☐ yes/是 ☑ no/否

If impact others product, Whether others projects&product change or not? 如果影响, 别的部件和产品是否也需要同时变更?

Design / 设计工程师 Signature / 签名:	卢清松	Date / 日期:	2026.3.10
Check / 校对工程师 Signature / 签名:	杨拓	Date / 日期:	2026.03.25

Description of Change/更改描述: (下面空白处请列出详细的变更前后对比)

具体变更详见附件PPT



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Design Change Review Sheet
设计变更评审表

Version 13
Effective date: 2026-01-14

Design Change Request No. 26_001

1. Function&Performance will be influenced? / 系统性能影响? ☐ yes/是 ☐ no/否

If Yes, System impaction evaluation? 如果是，系统影响评估? (如果是初次样件放行，空白处请说明样件目的)		System/签名 沈伟波(张汉泉代签)
Whether need sample to test? Plan? 是否需要样件进行验证? 计划?		
Whether the catalyst information is confirmed? 催化剂信息是否确认? (物料号、图纸、SPEC)		Catalyst/签名 张汉泉

2. Interface&boundary (internal and customer) will be influenced? / 接口&边界(内部和客户)影响? ☑ yes/是 ☐ no/否

If impact custmer ,whether get confirm from customer side? 请说明影响和是否得到客户的确认?		Design /签名 卢清松

3. Mechanical strength&Durability will be influenced? / 机械强度&可靠性影响 ? ☐ yes/是 ☐ no/否

If Yes, How to evaluate? Plan? 如果是，请说明影响和如何评估,计划? (例如:仿真分析等)	结构强度风险评估	李年
Whether need sample to test? Plan? 是否需要样件进行验证? 计划?		Validation/签名

4. Product document influenced?

4.1 Interface FMEA-relevant / IFMEA:	☑ no/否 ☐ yes/是 Note	Design /签名 卢清松
4.2. Product FMEA-relevant / DFMEA:	☑ no/否 ☐ yes/是 Note	
4.3. Special Charecteristics /PSC: (针对C Sample)	☑ no/否 ☐ yes/是 Note	
4.4. IMDS relevant: (针对C Sample)	☑ no/否 ☐ yes/是 Note	
4.5 Offer drawing relevant:(针对C Sample)	☑ no/否 ☐ yes/是 Note	
4.6. TCD relevant:	☑ no/否 ☐ yes/是 Note	

5. Changed parts Stock& treatment / 需要变更零件的库存统计和处理?
(只针对涉及到已经生产过的零件变更，新放行零件和本条不相关)
☐ NA.(新零件放行,不涉及库存)
☐ 涉及到已经生产过的零件变更

RBCW RDC(外库)和RBCW 仓库，是否有库存，	☐ 无库存 ☐ 有库存, 库存数量	
供应商处(已做好，未发货)是否有库存，	☐ 无库存 ☐ 有库存, 库存数量	
LPN工厂是否有库存，	☐ 无库存 ☐ 有库存, 库存数量	
在途，运输中是否有库存	☐ 无库存 ☐ 有库存, 库存数量	PPS /签名 李英芳
被影响到的实物库存处理指示: (设计工程师组织会议讨论)	☐ 可以继续使用完 ☐ 改制后使用 ☐ 报废	Design /签名
	☐ 报废成本 改制费用 ; ☐	PPS/签名
已装车件处理方法（发客户或标定车等）(SMP组织会议讨论)	☐ 改制 ; 更换 ; ☐ 其他（单独说明）;	PM/签名

6. Influence on supplier part?

Was this change confirmed with Purchasing (Supplier) for feasibility? Feedback from Supplier?
变更后的零件，生产可行性供应商是否确认? 反馈?

工装更改，检具更改

Influence on purchasing part delivery time? 对采购件交货时间的影响?

涉及到变更的零件，博世已出PR，供应商还未来得及生产的订单，是否已经通知供应商取消. ☐ no/否 ☐ yes/是

Does the change influence on purchasing cost ? 变更对采购成本的影响?
☐ increase/增加,金额 +36 ☐ decrease/降低 ,金额 ☐ no change/不变

PPS/签名 李英芳

7. DFMA & Sample prodcution OPL?

必要时附加DFMA 文件

ZH DFMA

Design /签名 卢清松

8. Influence on product manufacturing & assembly process?

Feedback from manufacturing engineer? 生产制造装配是否影响，反馈?

工装更改

PUQ/签名 [illegible]

Influence on project&product delivery time? 对项目&产品交货时间的影响? (795)

Influence on manufacturing cost / 对生产成本的影响? 在JIM493 2026 现有价格 +36
☐ increase/增加 ,金额 ☐ decrease/降低 ,金额 ☐ no change/不变

(33 CNY 达芬 +3 CNY 分摊) PPS/签名 李英芳

9. PM alignment? (Influence on project Timeline? Cost?);
是否同意对项目时间和成本的影响 ; 需求样品订单数量 ; 客户端 涨价 55 CNY (40 CNY 达芬 +150 CNY 分摊) PM/签名 [illegible]

ENG-ATS	PPS	PUQ	TCR3
Signature / 签名:	Signature / 签名:	Signature / 签名: Xue Tony	Signature / 签名:

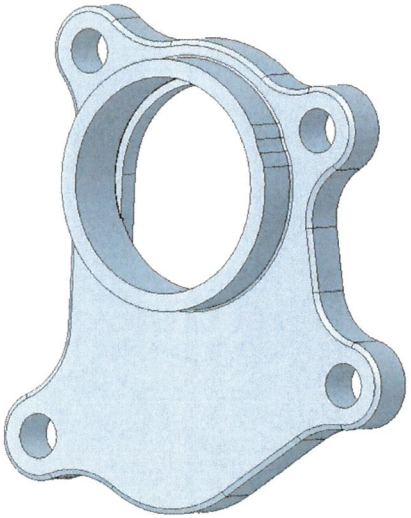
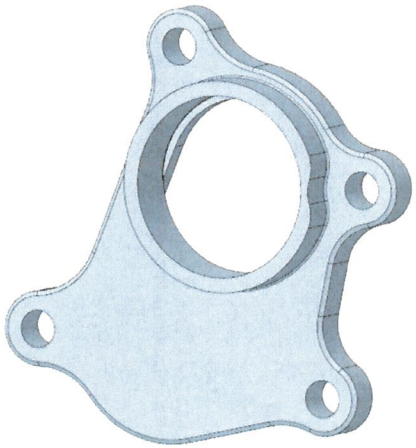
2026.3.9

PDECR26-001

ENG-CV

BOSCH

PDECR 26-001

	零件号	零件名称	Project	样件阶段	变更来源
	F01Z51039D-02	INLET FLANGE	JIM PT611	B C	BOSCH/Customer
变更描述	变更前		变更后		
	 F01Z51039D-00		 F01Z51039D-02		
变更原因	基于客户边界法兰数模调整，改为304材质的铸件法兰；				
风险点（影响）	凸台强度不足，有断裂风险；				
验证方法	FEA仿真，仿真报告见附件；				

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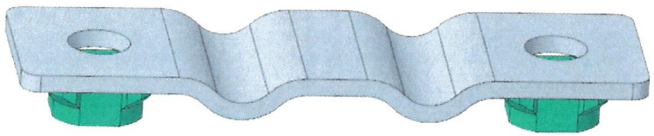
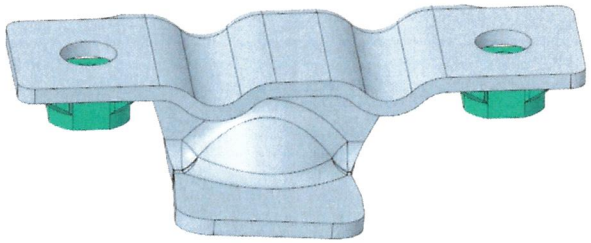
	零件号	零件名称	Project	样件阶段	变更来源
	See blow	BRACKET	JIM PT611	~~B~~ C	BOSCH/Customer

	变更前	变更后
变更描述	 BRACKET：F01Z5016N0 BRACKET：F01Z501103 NUT：F01Z5101CC	 BRACKET：F01Z50196V-00 BRACKET：F01Z510392-00 NUT：F01Z5101CC
变更原因	为方便供应商安装压差管定位，压差管夹片改为焊接支架；	
风险点（影响）	支架强度不足有断裂风险；	
验证方法	FEA仿真，仿真报告见附件；	

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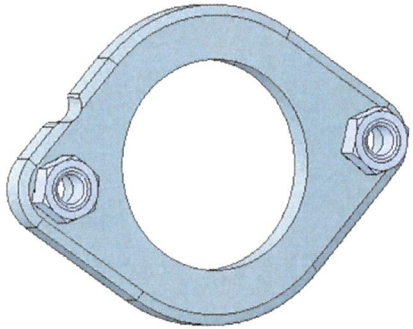
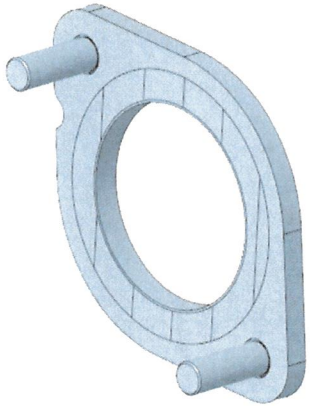
	零件号	零件名称	Project	样件阶段	变更来源
	See blow	FLANGE	JIM PT611	~~B~~C	BOSCH/Customer

	变更前	变更后
变更描述	 FLANGE：F01Z5017XS FLANGE：F01Z5102ET NUT：F01Z500753	 FLANGE：F01Z50199H-00 FLANGE：F01Z5101DL NUT：F01Z5103AE-00
变更原因	PT611与493和高扭中兴主项目统一，DPF出气法兰；	
风险点（影响）	支架强度不足有断裂风险；	
验证方法	FEA仿真，仿真报告见附件；	

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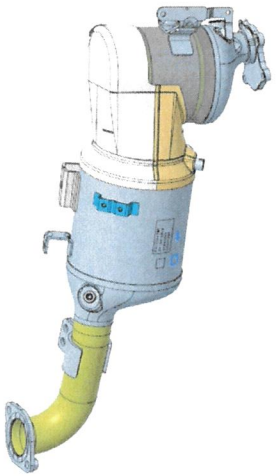
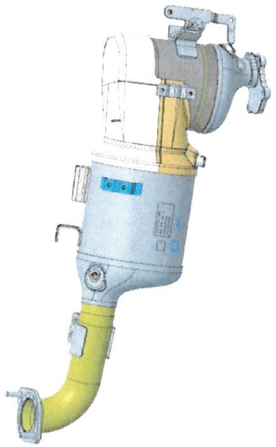
	零件号	零件名称	Project	样件阶段	变更来源
	F01Z50197F-01	DOC+SDPF WELDING	JIM PT611	B C	BOSCH/Customer

	变更前	变更后
变更描述	 F01Z50197F-00	 F01Z50197F-01
变更原因	上述变更后DOC+SDPF焊接总成随之改变;	
风险点（影响）	无	
验证方法	无	

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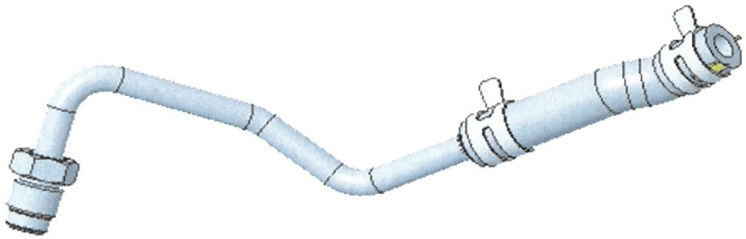
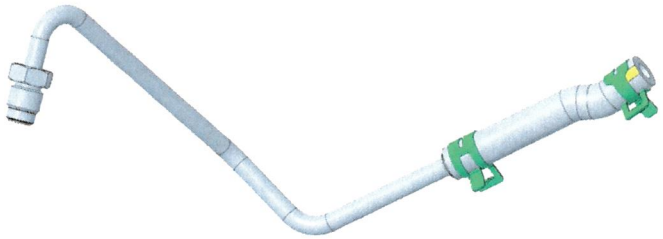
	零件号	零件名称	Project	样件阶段	变更来源
	See blow	DP tube inlet	JIM PT611	[redacted]	BOSCH/Customer

	变更前	变更后
变更描述	 DP tube inlet：F01Z5017W8 DP tube inlet：F01Z5017YK DP tube inlet：F01Z5102D5 RUBBER HOSE INLET：F01Z5102SV	 DP tube inlet：F01Z5019BS-00 DP tube inlet：F01Z5019BT-00 DP tube inlet：F01Z5103J3-00 RUBBER HOSE INLET：F01Z5103J4-00
变更原因	压差管夹片改为焊接支架后，压差硬管加长15mm，压差软管减短15mm；	
风险点（影响）	无	
验证方法	无	

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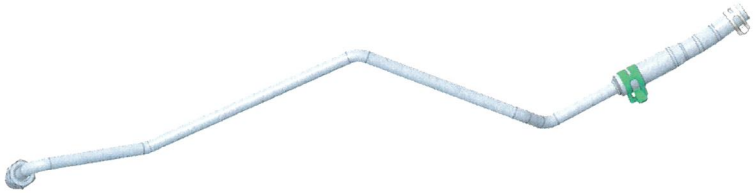
	零件号	零件名称	Project	样件阶段	变更来源
	See blow	DP tube inlet	JIM PT611	B C	BOSCH/Customer

	变更前	变更后
变更描述	 DP tube inlet：F01Z5017W7 DP tube inlet：F01Z5017YJ DP tube inlet：F01Z5102D4 RUBBER HOSE INLET：F01Z5102SW	 DP tube inlet：F01Z5019BU-00 DP tube inlet：F01Z5019BV-00 DP tube inlet：F01Z5103J5-00 RUBBER HOSE INLET：F01Z5103J6-00
变更原因	压差管夹片改为焊接支架后，压差硬管加长15mm，压差软管减短15mm；	
风险点（影响）	无	
验证方法	无	

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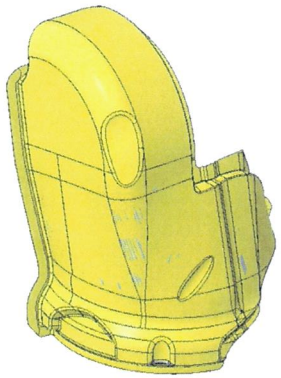
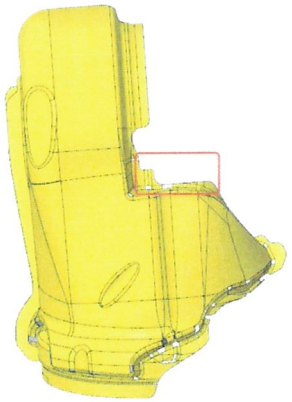
	零件号	零件名称	Project	样件阶段	变更来源
	See blow	MIXER INSULATION 02	JIM PT611	~~B~~C	BOSCH/Customer

	变更前	变更后
变更描述	 INSULATION：F01Z5018AY FOIL：F01Z5017NU MAT：F01Z5102RM MAT：F01Z5102RN	 INSULATION：F01Z5019BX-00 FOIL：F01Z5103J9-00 MAT：F01Z5102RM MAT：F01Z5102RN
变更原因	压差支架改焊接后，保温壳增加避让区域；	
风险点（影响）	无	
验证方法	无	

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	零件号	零件名称	Project	样件阶段	变更来源
	See blow	INSULATION	JIM PT611	~~B~~C	BOSCH/Customer

	变更前	变更后
变更描述	 INSULATION：F01Z50197G-00 FOIL：F01Z5102RH MAT：F01Z5102RJ INSULATION：F01Z50197H-00 FOIL：F01Z5102DD MAT：F01Z5102RP MAT：F01Z5102RR	 INSULATION：F01Z50197G-00 FOIL：F01Z5102RH MAT：F01Z5102RJ INSULATION：F01Z50197H-01 FOIL：F01Z5102DD MAT：F01Z5102RP MAT：F01Z5102RR
变更原因	上述更改后，保温铭牌信息随之更改；	
风险点（影响）	无	
验证方法	无	

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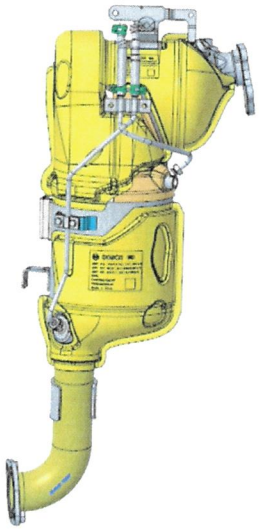
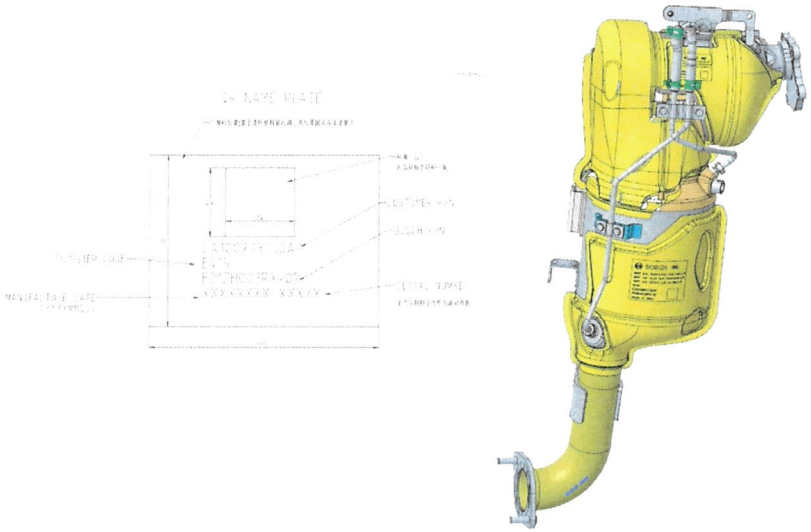
PDECR 26-001

	零件号	零件名称	Project	样件阶段	变更来源
	F01ZH003RX-02	DOC+SDPF UNIT	JIM PT611	~~B~~C	BOSCH/Customer
变更描述	变更前		变更后		
	 F01ZH003RX-01		 F01ZH003RX-02		
变更原因	1.添加纸质标签；2.上述更改后，DOC+SDPF总成随之更改；				
风险点（影响）	无				
验证方法	无				

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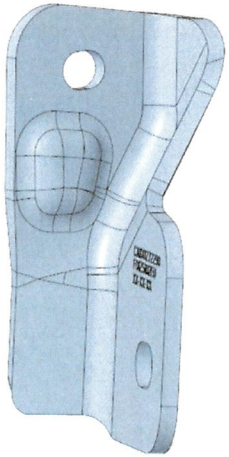
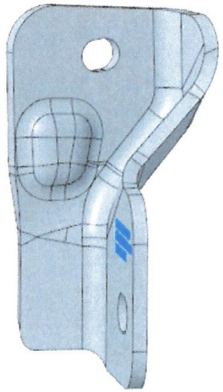
	零件号	零件名称	Project	样件阶段	变更来源
	F01Z5103J8-00	BRACKET	JIM PT611	~~B~~ C	BOSCH/Customer

	变更前	变更后
变更描述	 F01Z5102EU	 F01Z5103J8-00
变更原因	客户边界更改后，支架的形状及安装孔位随之更改；	
风险点（影响）	无	
验证方法	无	

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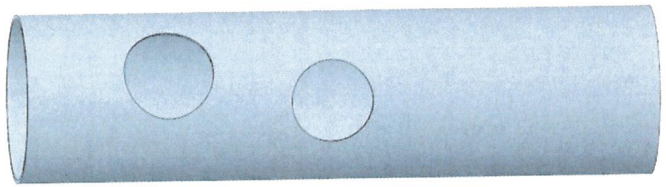
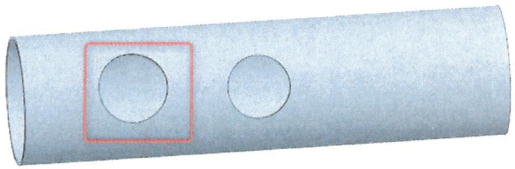
PDECR 26-001

	零件号	零件名称	Project	样件阶段	变更来源
	F01Z5103J7-00	OUTLET PIPE	JIM PT611	B C	BOSCH/Customer

	变更前	变更后
变更描述	 F01Z5101D5	 F01Z5103J7-00
变更原因	客户要求NOX座向下旋转15°，SCR尾管NOX座孔位随之更改；	
风险点（影响）	无	
验证方法	无	

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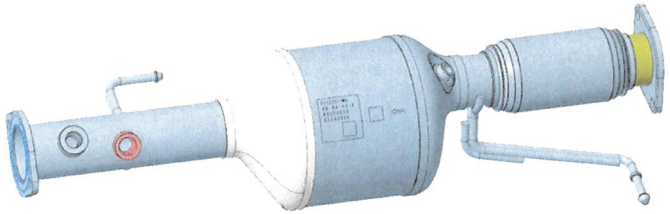
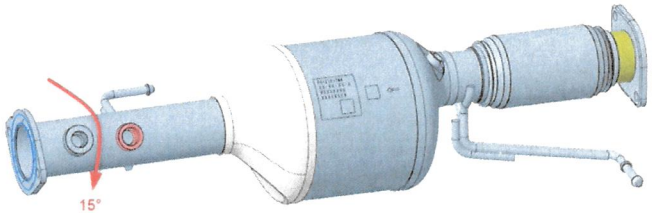
PDECR 26-001

	零件号	零件名称	Project	样件阶段	变更来源
	F01Z5019BW-00	SCR WELDING	JIM PT611	BC	BOSCH/Customer
变更描述	变更前		变更后		
	 F01Z5017W9		 F01Z5019BW-00		
变更原因	客户要求NOX座向下旋转15°，SCR尾管NOX座孔位随之更改；				
风险点（影响）	无				
验证方法	无				

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PDECR 26-001

	零件号	零件名称	Project	样件阶段	变更来源
	See blow	INSULATION	JIM PT611	~~B~~ C	BOSCH/Customer

	变更前	变更后
变更描述	 INSULATION：F01Z5018C0 FOIL：F01Z5102T8 MAT：F01Z51018D	 INSULATION：F01Z5019BY-00 FOIL：F01Z5102T8 MAT：F01Z51018D
变更原因	上述更改后，SCR保温铭牌信息随之更改；	
风险点（影响）	无	
验证方法	无	

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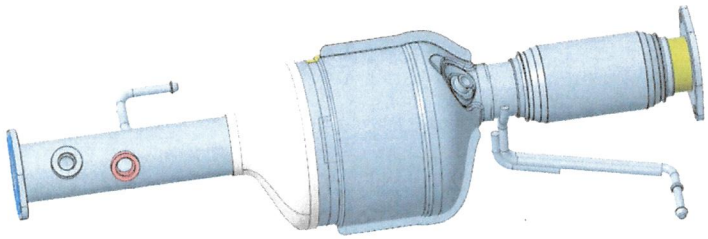
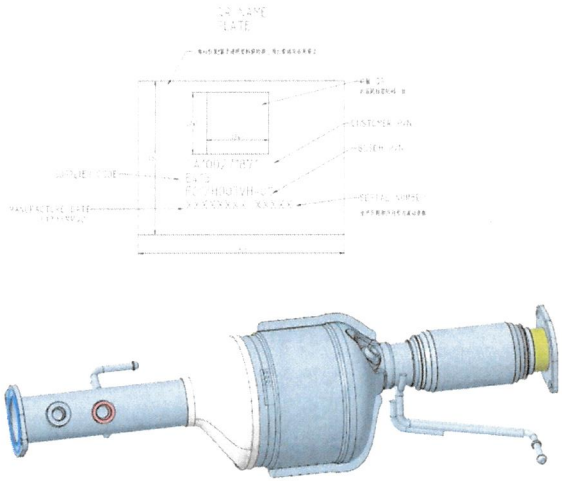
PDECR 26-001

	零件号	零件名称	Project	样件阶段	变更来源
	F01ZH003VH-00	SCR UNIT	JIM PT611	~~B~~ C	BOSCH/Customer

	变更前	变更后
变更描述	 F01ZH003FV	 F01ZH003VH-00
变更原因	1.添加纸质标签；2.上述更改后，SCR总成随之更改；	
风险点（影响）	无	
验证方法	无	

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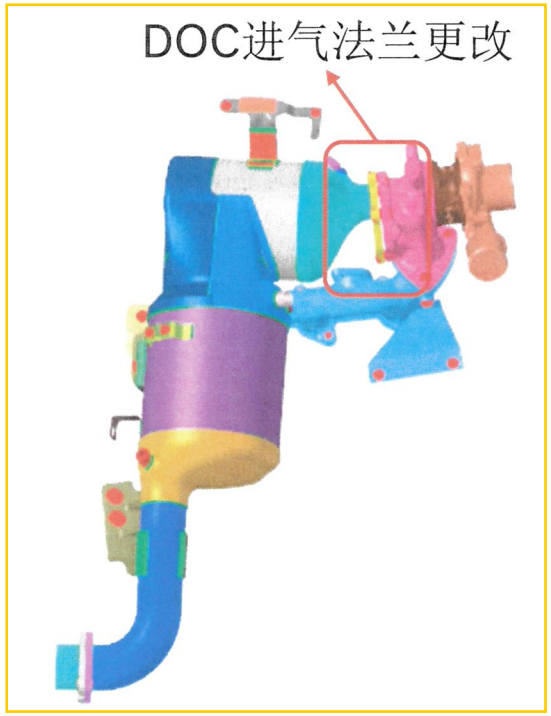
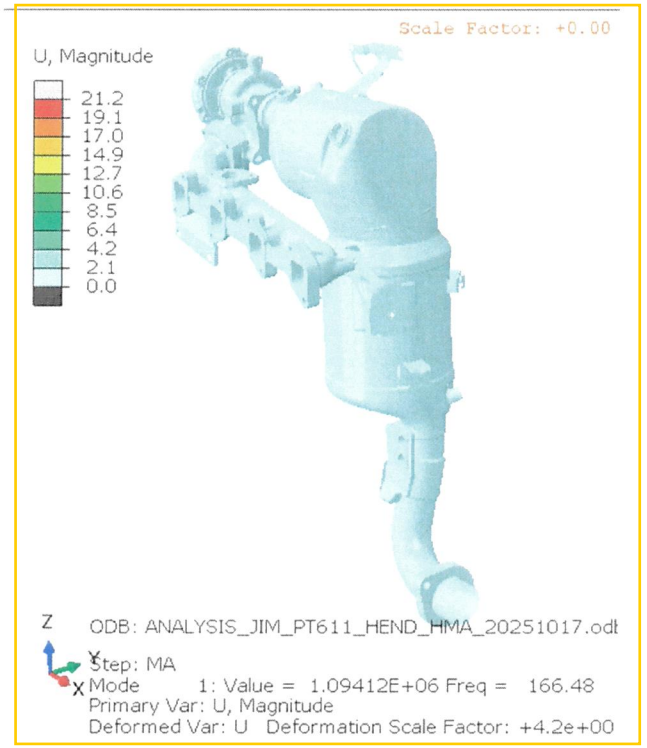
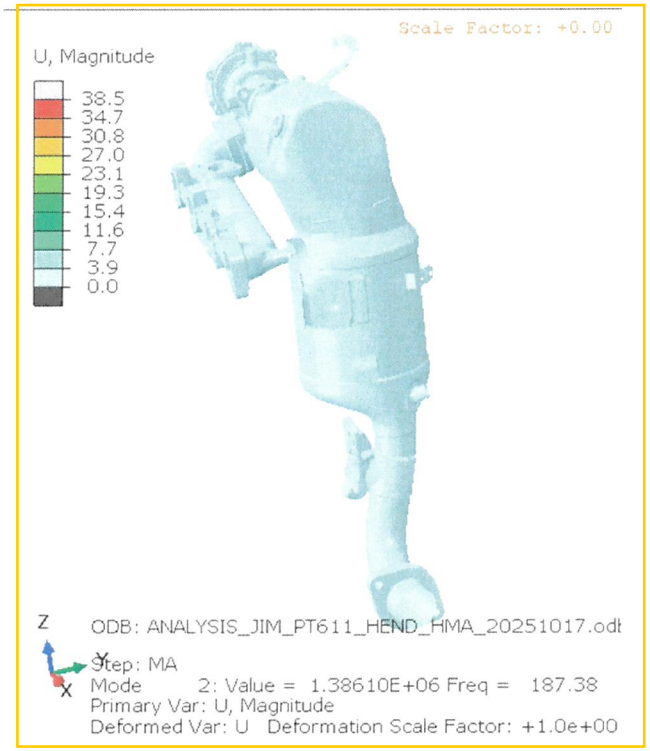
JIM_PT611_HOTEND_HMA

两支架（冲压）-频率

LIMIT＞150Hz







一阶模态频率为166.5Hz。

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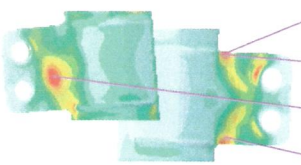
JIM_PT611_HOTEND_FRA（5g频响）

两支架（冲压）-应力

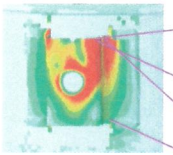
Ford Limit :

Temp	409 Non-weld Area	409 Weld Area	441 Non-weld Area	441 Weld Area
300	110	69	125	78
400	85	53	115	72
500	72	43	102	61
600	50	30	89	53
700	30	18	65	39

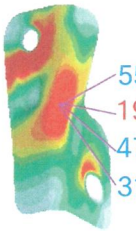
母材应力：limit 110MPa（300℃）



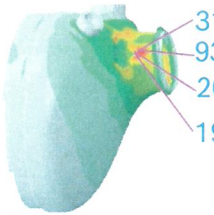
37.1MPa@ 150Hz(X)
103.7MPa@ 166.5Hz(X)
111.3MPa@ 187.4Hz(Z)
32.1MPa@ 200Hz(Z)



68.1MPa@ 150Hz(X)
235.8MPa@ 166.5Hz(X)
35.5MPa@ 200Hz(X)
78.6MPa@ 187.4Hz(Z)

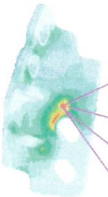


55.1MPa@ 150Hz(X)
195.8MPa@ 166.5Hz(X)
47.8MPa@ 187.4Hz(X)
31.3MPa@ 200Hz(X)



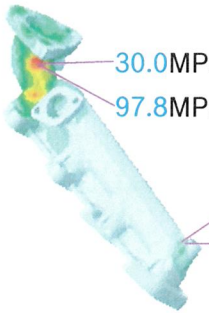
31.7MPa@ 150Hz(X)
93.1MPa@ 166.5Hz(X)
20.8MPa@ 187.4Hz(Y)
19.2MPa@ 200Hz(Y)

铸件应力： limit 165MPa（180℃）



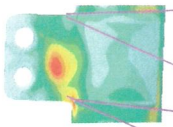
39.5MPa@ 150Hz(X)
118.5MPa@166.5Hz(X)
170.1MPa@187.4Hz(Z)
68.9MPa@ 200Hz(Z)

铸件应力： limit 118MPa（400℃）

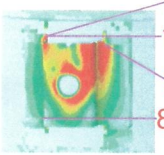


30.0MPa@ 150Hz(X)
97.8MPa@ 166.5Hz(X)
24.8MPa@ 187.4Hz(Z)
19.8MPa@ 200Hz(Y)

焊缝应力：limit 69MPa（300℃）

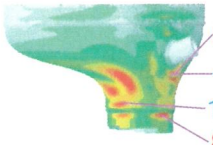


21.0MPa@ 150Hz(X)
52.6MPa@ 166.5Hz(X)
107.7MPa@ 187.4Hz(Z)
44.3MPa@ 200Hz(Y)



63.0MPa@ 150Hz(X)
195.3MPa@ 166.5Hz(X)
27.7MPa@ 200Hz(X)
83.5MPa@ 187.4Hz(Z)

21.9MPa@ 150Hz(X)
64.7MPa@ 166.5Hz(X)
20.1MPa@ 187.4Hz(Y)
21.1MPa@ 200Hz(Y)



53.3MPa@ 200Hz(Y)
46.6MPa@ 166.5Hz(X)
16.5MPa@ 150Hz(X)
80.4MPa@ 187.4Hz(Z)

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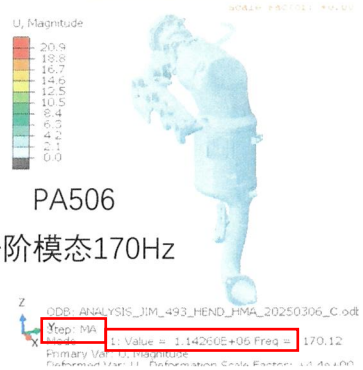
FEA结果对比

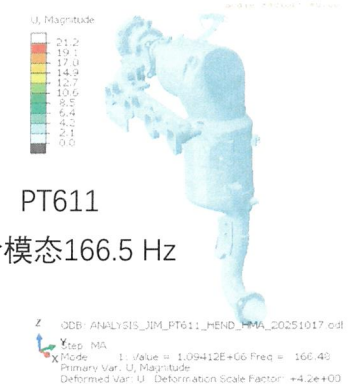
PT611 vs PA506（493主项目）

要求>150Hz



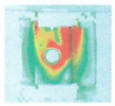
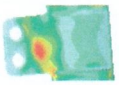
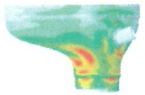
PA506

一阶模态170Hz



PT611

一阶模态166.5 Hz

	5g频响应力@150Hz；单位 Mpa，限值110Mpa					
	支架1	支架2	支架3	支架4	进气端锥	出气端锥
				限值165Mpa		
PA506	59.3	41	32.7	36.4	26.7	15.6
PT611	68.1	55.1	37.1	39.5	31.7	16.5

相比PA506，PT611模态频率有所下降，支架端锥应力有所上升，单仍然满足限值要求，风险较低

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